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PROJECT – PART B

Simulation and Analysis of the Output Characteristics of a BJT Common-Base Amplifier

Question 2b: BJT as a Current-Controlled Current Source (CCCS) and CB-BJT as a Current Buffer

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1. Problem statement

Q. No.	Problem Description
2b	Simulate, observe and analyze the output characteristics of a BJT common base amplifier, and explain the following: **(i) BJT is called a current controlled current source; (ii) CB-BJT can be used as current buffers.**

*The two parts of the question are answered separately and are highlighted in **Section 7** (answer to i) and **Section 8** (answer to ii). Sections 3-6 build the evidence they rely on.*

2. Aim and objectives

The aim of this work is to study a Bipolar Junction Transistor (BJT) connected in the common-base (CB) configuration, to obtain and interpret its output characteristics, and to use those characteristics to justify two statements: a BJT is a current-controlled current source, and a common-base stage can act as a current buffer.

- Draw the CB amplifier and the test circuit used to obtain the output characteristics.
- Plot I_C versus V_{CB} for several fixed values of the emitter current I_E and identify the saturation, active and cutoff regions.
- Show from the data that $I_C \approx \alpha \cdot I_E$ and is almost independent of V_{CB} (current-controlled current source).
- Show that the very low input resistance and very high output resistance of the CB stage make it a current buffer, and demonstrate this with a worked example.

3. Theory: BJT and the common-base configuration

3.1 The bipolar junction transistor

A BJT is a three-terminal device made of two back-to-back p-n junctions: the emitter (heavily doped, supplies carriers), the base (very thin and lightly doped) and the collector (collects the carriers). For an NPN transistor in normal (forward-active) operation the emitter-base junction is forward biased and the collector-base junction is reverse biased. Electrons injected from the emitter cross the thin base and are swept into the collector by the reverse-biased junction; only a small fraction recombines in the base and appears as base current.

The terminal currents obey Kirchhoff's current law:

$$I_E = I_B + I_C$$

3.2 Common-base current gain (α)

In the common-base configuration the base is the terminal common to the input (emitter-base) and the output (collector-base) loops. The input current is I_E and the output current is I_C. The ratio of the two is the common-base current gain:

$$\alpha = I_C / I_E \text{ (typically 0.95 – 0.995, always slightly less than 1)}$$

Including the small reverse leakage current of the collector-base junction (I_{CBO}, the collector current when $I_E = 0$):

$$I_C = \alpha \cdot I_E + I_{CBO}$$

α is related to the common-emitter gain β by $\beta = \alpha/(1 - \alpha)$. For $\alpha = 0.99$ this gives $\beta \approx 99$.

3.3 The three regions of operation

Region	Emitter-base junction	Collector-base junction	Behaviour in CB
Active	Forward biased	Reverse biased ($V_{CB} > 0$)	$I_C \approx \alpha \cdot I_E$; flat curves; used for amplification
Saturation	Forward biased	Forward biased ($V_{CB} < 0$)	I_C falls rapidly to zero as V_{CB} becomes negative (≈ -0.5 V and below)
Cutoff	Reverse biased ($I_E = 0$)	Reverse biased	$I_C \approx I_{CBO}$ (nanoamps) – transistor is OFF

3.4 Why the output curves are flat

Once V_{CB} is positive the collector-base junction is reverse biased and collects practically every carrier that reaches it. The number of carriers reaching it is decided by the emitter injection, i.e. by I_E, not by how large V_{CB} is. Increasing V_{CB} only widens the depletion region slightly (the Early effect), which makes I_C rise by a very small amount. This is why the output characteristics are almost horizontal lines – the central fact behind both parts of the question.

4. Circuit diagram and simulation set-up

Figure 1 shows a complete common-base amplifier. The base is held at AC ground; the signal is applied at the emitter through coupling capacitor C_1 and the amplified signal is taken from the collector through C_2. R_E sets the emitter bias current from the negative supply $-V_{EE}$ and R_C develops the output voltage from the supply $+V_{CC}$.

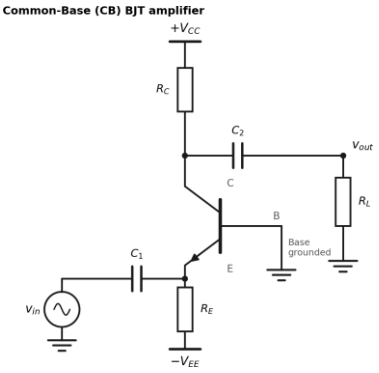


Figure 1: Common-base BJT amplifier (NPN) with coupling capacitors, bias resistors and load.

To obtain the *output characteristics* the emitter is driven by a constant current source I_E and the collector–base voltage V_{CB} is swept with a variable supply while the collector current is read on an ammeter (Figure 2). I_E is then stepped (0, 1, 2, 3, 4, 5 mA) and the sweep is repeated for each value.

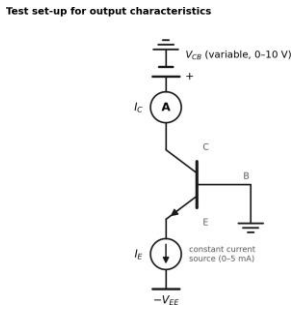


Figure 2: Test set-up used to plot I_C versus V_{CB} at constant I_E.

Components and settings

Component	Value / type	Purpose
Q1	NPN BJT (e.g. 2N2222 / BC547)	Device under study
V_{CC}	+10 V to +12 V	Collector supply
V_{EE}	−10 V	Emitter bias supply
R_E	≈ 9.3 kΩ (for $I_E \approx 1$ mA)	Sets emitter bias current
R_C	2.2 kΩ	Collector load
R_L	10 kΩ	External load
C_1, C_2	10 µF	Coupling capacitors
v_{in}	5 mV peak, 1 kHz sine	Small input signal
Constant current source	0 – 5 mA	Fixes I_E for the characteristics
V_{CB} sweep	−1 V to +10 V	Spans saturation and active regions

5. Simulation procedure

1. Open the circuit simulator (Tinkercad Circuits, Multisim, LTspice or similar) and place an NPN transistor.
2. Ground the base. Connect a constant current source (or a resistor from $-V_{EE}$) to the emitter and a variable DC supply V_{CB} between the collector and ground through an ammeter, as in Figure 2.
3. Set $I_E = 1$ mA. Sweep V_{CB} from -1 V to 10 V and record I_C at each step.
4. Repeat for $I_E = 2$, 3, 4 and 5 mA, and also for $I_E = 0$ mA (cutoff).
5. Plot I_C (y-axis) against V_{CB} (x-axis) for all curves on one graph – the output characteristics.
6. For the amplifier of Figure 1, apply a 5 mV, 1 kHz sine at the emitter and observe v_{in} and v_{out} on the oscilloscope.
7. Compute $\alpha = I_C / I_E$ from the readings and compare with the theoretical value.

Note on the plots: the graphs in this report are generated from a standard first-order BJT model ($\alpha_0 = 0.99$, Early-type slope $V_A = 2000$ V, $I_{CBO} = 20$ nA, $V_T \approx 26$ mV). A real simulator will give slightly different numbers for a particular transistor model, but the shape and conclusions are the same.

6. Observations and output characteristics

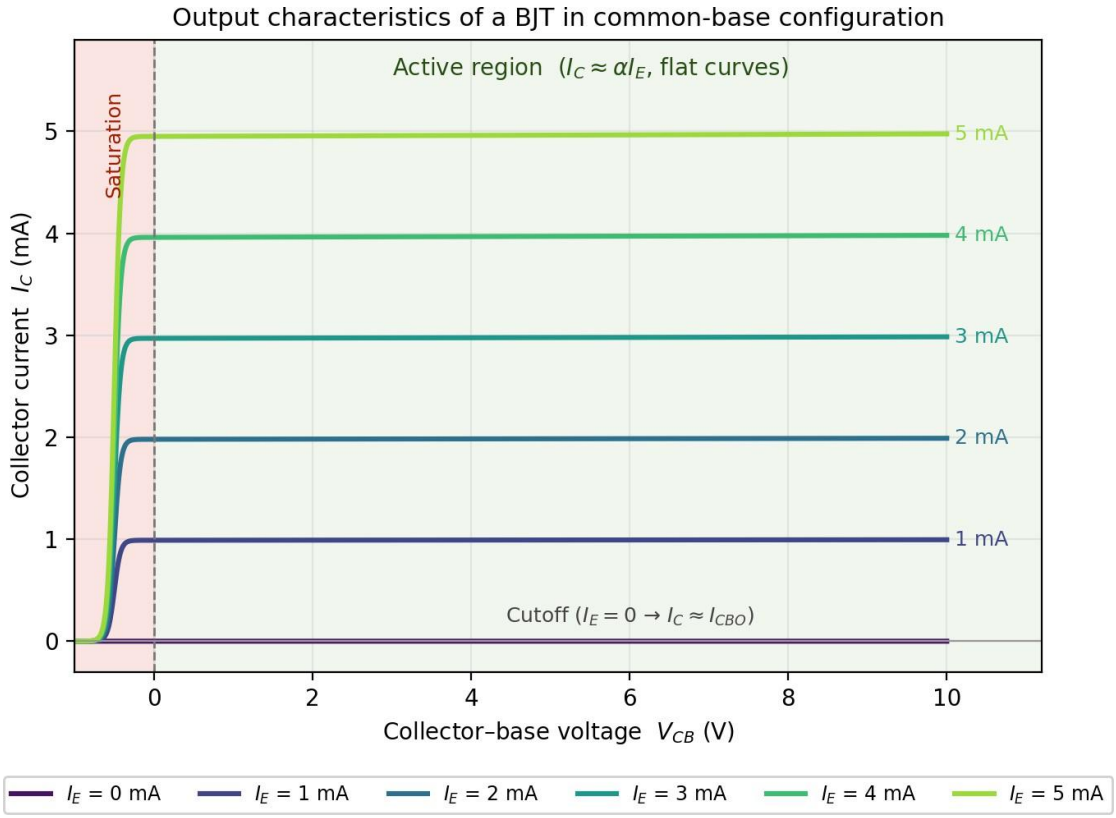


Figure 3: Output characteristics I_C vs V_{CB} for I_E = 0 to 5 mA (common-base).

What the graph shows.

- **Active region (V_{CB} > 0):** each curve is almost horizontal. I_C stays close to $\alpha \cdot I_E$ across the whole 0–10 V range.
- **Spacing between curves:** equal steps of I_E give equal steps of I_C – I_C is proportional to I_E.
- **Saturation region (V_{CB} < 0):** the collector–base junction becomes forward biased and I_C collapses within a few hundred millivolts below zero.
- **Cutoff (I_E = 0):** the bottom curve sits on the axis; only the tiny leakage I_{CBO} flows.

Tabulated readings

Table 1 – I_C for different emitter currents at fixed V_{CB} = 5 V:

I_E (mA)	I_C (mA) at V_{CB} = 5 V	$\alpha = I_C/I_E$	$I_B = I_E - I_C$ (µA)
1.0	0.992	0.9925	7.5
2.0	1.985	0.9925	15.1
3.0	2.977	0.9925	22.6
4.0	3.970	0.9925	30.1
5.0	4.962	0.9925	37.6

The ratio I_C/I_E is the same (≈ 0.99) at every emitter current, confirming that the collector current is a fixed fraction of the emitter current.

Table 2 – effect of V_{CB} on I_C at fixed I_E = 2 mA:

V_{CB} (V)	0	2	4	6	8	10
I_C (mA) at I_E = 2 mA	1.9800	1.9820	1.9840	1.9859	1.9879	1.9899

Changing V_{CB} from 0 V to 10 V (a 10 V change) alters I_C by only about 0.010 mA (0.5%). The slope of the curves gives the output resistance of the stage, $r_o = \Delta V_{CB}/\Delta I_C \approx 1.0$ MΩ at I_E = 2 mA – extremely high, as expected of a good current source.

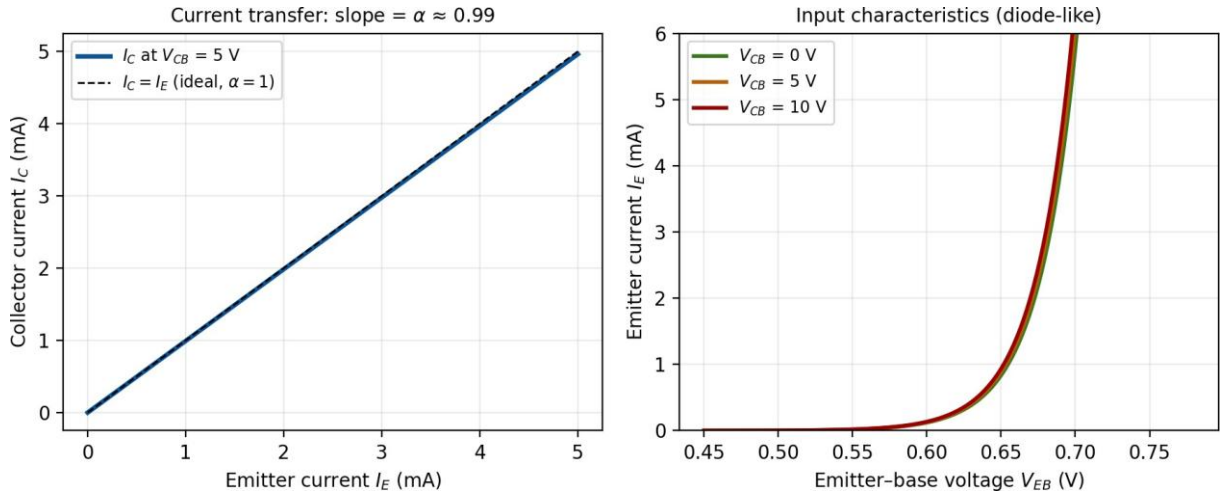


Figure 4: (left) Transfer characteristic I_C vs I_E - a straight line of slope α. (right) Input characteristics - a diode-like exponential.

ANSWER TO QUESTION (i)

7. The BJT is a Current-Controlled Current Source

Q 2b(i): Explain why a BJT is called a current controlled current source.

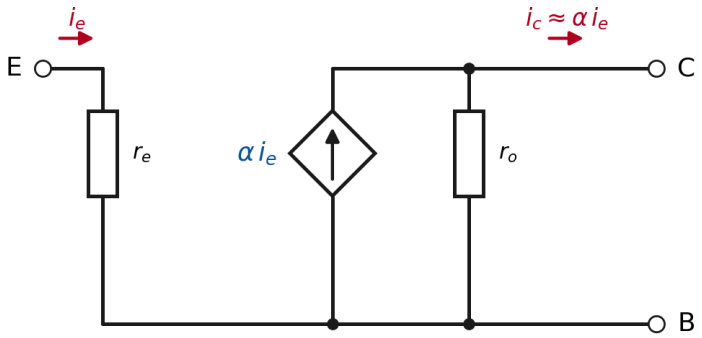
Short answer

In the common-base connection the collector (output) current is set by the emitter (input) *current*, $I_C = \alpha \cdot I_E$, and it stays almost the same whatever the output voltage V_{CB} is ($r_o \approx 1$ MΩ). An output current that is a fixed multiple of a controlling current, and independent of the output voltage, is exactly a **current-controlled current source.**

A **current-controlled current source (CCCS)** is an element whose output current is a constant multiple of a controlling current elsewhere in the circuit and does not depend on the voltage across the output terminals. The CB characteristics show exactly this behaviour.

7.1 Evidence from the characteristics

- **Controlled by current:** the collector (output) current is set by the emitter (input) current. Stepping I_E in equal increments moves I_C in equal increments (Figure 3, Figure 4-left): $I_C = \alpha \cdot I_E$.
- **Behaves as a source:** for a given I_E the output current is nearly independent of the output voltage V_{CB} (flat curves, Table 2). A source delivers the same current into any load.
- **Very high output resistance:** the curves are almost horizontal, so r_o is in the MΩ range – the hallmark of an ideal current source (an ideal one has $r_o = \infty$).



Small-signal CB model: output current is a controlled copy of the input current

Figure 5: Small-signal model of the CB-BJT. The dependent source $a \cdot i_e$ is controlled by the emitter current; r_o is its (large) output resistance.

7.2 Same idea in the common-emitter view

The same device is also a CCCS when the base current is regarded as the control: $I_C = \beta \cdot I_B$ with $\beta = \alpha/(1-\alpha)$. A small base current controls a β-times larger collector current. Whichever terminal is used as the input, the collector current is *produced by a current* and is *insensitive to the collector voltage*, so the BJT is a CCCS. (This differs from a MOSFET, which is a voltage-controlled current source.)

7.3 Constant current into a changing load

Because I_C is fixed by I_E, changing the collector resistor does not change the collector current as long as the transistor stays in the active region. The load only changes the *voltage* developed (V =

$I_C \cdot R_C$). This is the defining property of a current source (Figure 6).

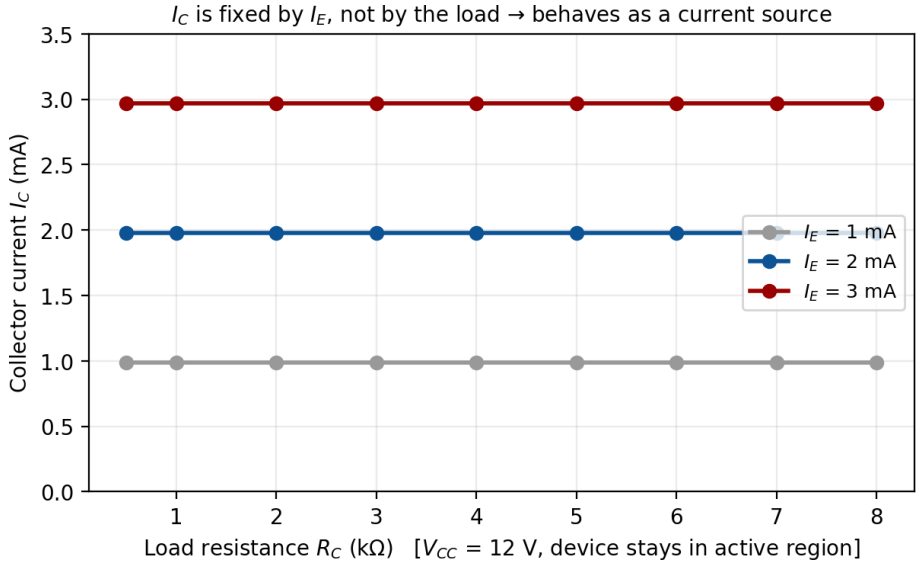


Figure 6: I_C stays constant while R_C is varied - the transistor behaves as a current source set by I_E.

7.4 Load line and operating point

When a collector resistor R_C is used, the operating point is where the load line meets the characteristic for the chosen I_E. For V_{CC} = 10 V and R_C = 2 kΩ the load line is $V_{CB} = V_{CC} - I_C R_C$, and with I_E = 2 mA the Q-point is $V_{CB} \approx$ 6.03 V, $I_C \approx$ 1.99 mA – safely in the active region, far from saturation.

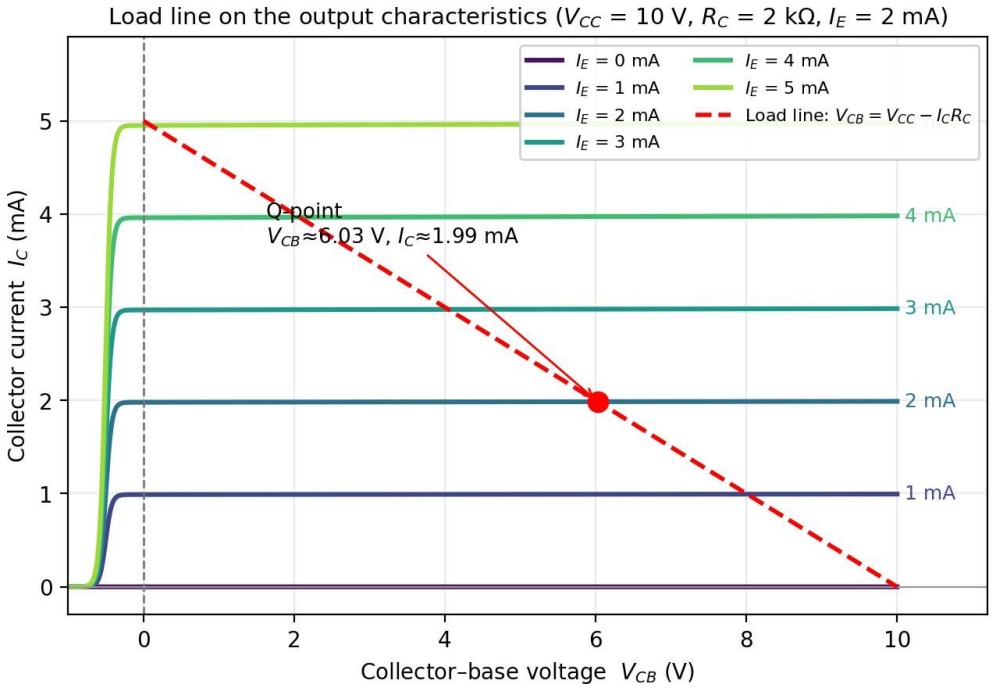


Figure 7: Load line (dashed red) and Q-point on the output characteristics.

Conclusion for (i)

$I_C \approx \alpha \cdot I_E$ is determined by a controlling current and is practically independent of the output voltage; therefore the BJT is correctly called a **current-controlled current source.**

ANSWER TO QUESTION (ii)

8. The CB-BJT as a Current Buffer

Q 2b(ii): Explain why a CB-BJT can be used as a current buffer.

Short answer

A current buffer should pass the input current to the load almost unchanged while isolating the source from the load. The CB stage has a very **low input resistance** ($r_e = V_T/I_E \approx 26\ \Omega$), a very **high output resistance** ($r_o \approx 1\ M\Omega$) and a **current gain α ≈ 1**. So it takes in nearly all of the source current and delivers it to the load independent of the load value – the behaviour of a **current buffer.**

A **current buffer** accepts a current at its input and delivers (almost) the same current at its output, while isolating the source from the load. An ideal current buffer has **zero input resistance** (so it absorbs the whole source current), **infinite output resistance** (so the output current does not depend on the load) and a **current gain of 1**. The CB stage comes very close to this.

8.1 Properties of the CB stage

Parameter	Expression	Typical value	Ideal current buffer
Input resistance r_{in}	$r_e = V_T/I_E$	26 Ω at 1 mA; 13 Ω at 2 mA	0
Output resistance r_{out}	r_o (in parallel with R_C)	≈ 1.0 MΩ (device)	∞
Current gain A_i	$i_c/i_e = \alpha$	0.99 (≈ 1)	1

The input resistance is small because the emitter–base junction is a forward-biased diode with $r_e = V_T/I_E$; the output resistance is large because the collector–base junction is reverse biased (flat curves). The current gain α is just under unity, so the current at the output is essentially a copy of the current at the input – it is *buffered*, not amplified.

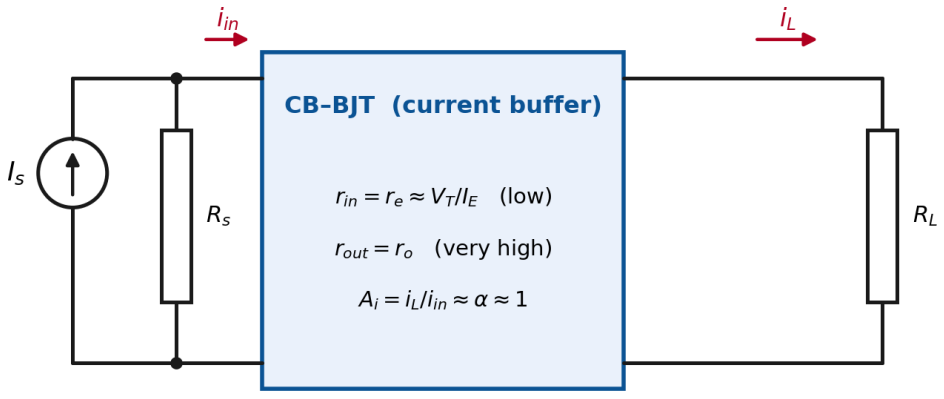


Figure 8: CB-BJT inserted between a current source (I_s with source resistance R_s) and a load R_L.

8.2 Worked example - why a buffer is needed

Take a signal source of $I_s = 1$ mA with source resistance $R_s = 1$ kΩ driving a load $R_L = 10$ kΩ.

Without the buffer the source current divides between R_s and R_L:

$$I_L = I_s \cdot R_s/(R_s+R_L) = 1\ \text{mA} \times 1\text{k}/(1\text{k}+10\text{k}) = \mathbf{0.091\ mA}\ \text{(only 9 \% reaches the load)}$$

With a CB-BJT buffer ($r_{in} = r_e = 26\ \Omega$ at $I_E = 1$ mA, $r_o = 1$ MΩ, $\alpha = 0.99$) the source now sees only 26 Ω:

$$i_{in} = I_s \cdot R_s/(R_s+r_e) = 0.975\ \text{mA},\ I_L = \alpha \cdot i_{in} \cdot r_o/(r_o+R_L) = \mathbf{0.955\ mA}$$

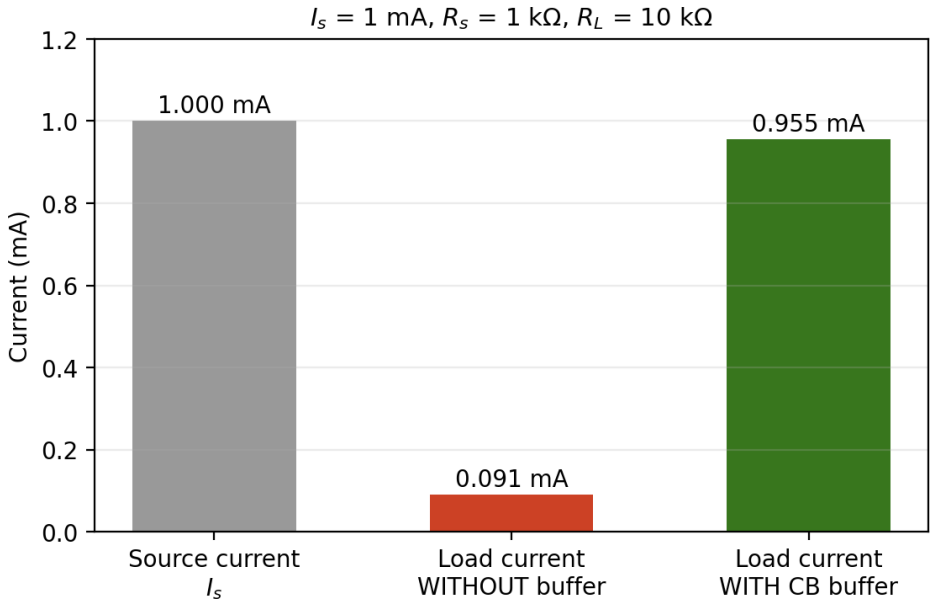


Figure 9: Load current with and without the CB current buffer (same source and load).

The buffer raises the load current from 0.091 mA to 0.955 mA, i.e. about 11 times more of the source current reaches the load. Because $r_o \gg R_L$, the delivered current is also independent of the load value.

8.3 Where the CB buffer is used

- **Cascode amplifiers:** a CB transistor on top of a common-emitter transistor passes on the collector current of the CE stage with a very low input resistance, which removes the Miller effect and greatly extends bandwidth.
- **High-frequency / RF amplifiers:** the CB stage has good frequency response and low input resistance, suited to low-impedance sources such as antennas and cables.
- **Current mirrors and current sources:** the output stage of precision current sources uses the high output resistance of the CB connection.
- **Interfacing:** connecting a current-output source (photodiode, sensor) to a load while isolating the two.

Conclusion for (ii)
With low r_{in}, very high r_{out} and $A_i \approx 1$, the CB-BJT transfers the input current to the load almost unchanged, which is exactly what a **current buffer** is required to do.

9. Small-signal behaviour of the CB amplifier

Although the CB stage is a current buffer ($A_i \approx 1$), it can still give a large *voltage* gain because a small emitter current change produces a collector current change that flows through a much larger resistor R_C.

$$A_v = v_{out}/v_{in} \approx g_m \cdot (R_C \parallel R_L) \approx \alpha \cdot R_C/r_e$$

With $I_E = 1$ mA, $g_m = I_C/V_T \approx 38$ mS and $R_C = 2.2$ kΩ (taking R_L large) the gain is about 84. For a 5 mV input the output swing is ≈ 421 mV peak (Figure 10). **The output is in phase with the input** – the CB stage does not invert.

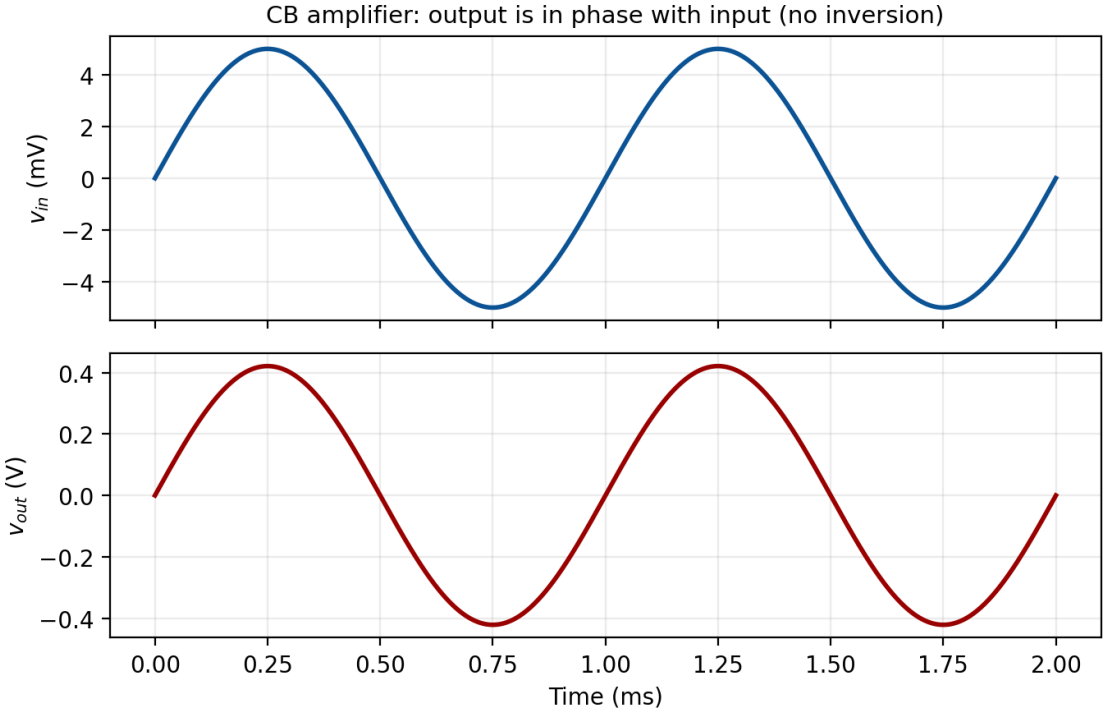


Figure 10: Input (top) and output (bottom) waveforms of the CB amplifier - same phase, larger amplitude.

Property	Common base (CB)
Input resistance	Very low (≈ r_e, tens of Ω)
Output resistance	Very high (≈ R_C in the circuit; r_o of the device is MΩ)
Current gain	$\alpha \approx 1$ (less than 1)
Voltage gain	High (≈ $g_m R_C$)
Phase relation	Output in phase with input (0°)
Frequency response	Excellent – widest bandwidth of the three configurations

10. Results, conclusion and applications

10.1 Summary of results

Quantity	Result
Common-base current gain α	≈ 0.99 (I_C/I_E, constant for all I_E)
Equivalent β	≈ 99
Change in I_C for 10 V change in V_{CB} (I_E = 2 mA)	≈ 0.5 %
Output resistance (device)	≈ 1.0 MΩ
Input resistance at I_E = 1 mA	≈ 26 Ω
Q-point (V_{CC} = 10 V, R_C = 2 kΩ, I_E = 2 mA)	V_{CB} ≈ 6.03 V, I_C ≈ 1.99 mA
Voltage gain (R_C = 2.2 kΩ, I_E = 1 mA)	≈ 84, non-inverting

10.2 Conclusion

The simulated output characteristics of the common-base BJT consist of nearly flat lines of I_C against V_{CB} in the active region, equally spaced for equal steps of I_E, with a steep fall into saturation for V_{CB} below zero and only leakage current at cutoff. These observations show that:

- **(i)** the collector current is controlled by the emitter current ($I_C = \alpha I_E$) and is almost independent of the output voltage with a very large output resistance – therefore the BJT is a **current-controlled current source**;
- **(ii)** the CB stage has very low input resistance, very high output resistance and a current gain close to unity – therefore it works as an excellent **current buffer**, as the worked example confirmed.

10.3 Precautions and sources of error

- Keep the transistor in the active region ($V_{CB} > 0$) when using it as a current source or buffer.
- Do not exceed the maximum collector current, collector-base voltage and power ratings of the device.
- Real transistors show temperature-dependent α and I_{CBO}; simulator models may differ slightly from the ideal model used for the plots.